

(Reprinted from Markkula Centre for Applied Ethics cases)

1. Conflict – of- interests case

Dylan is a manager at a leading consumer electronics company. In his role, he must frequently interact with industry partners and customers. One day, Dylan met with a customer of one of his company's custom products. The custom product was a special chip for an electronic appliance that was currently in its final stages of review before market release. During the meeting, the customer wanted to know the method of making the chip, a process which was not specified in the given datasheet. The client claimed this **information was needed to ensure that the chip would function properly when it was integrated with electronic appliances.**

At first, Dylan was uncertain. He wanted to give his customer more details if it was for the benefit of his client's final product, but, at the same time, was concerned because the requested information was protected under his company's non-disclosure agreement (NDA).

Dylan decided to discuss the matter with his supervisor; however, Dylan's manager was overseeing many projects and, knowing that Dylan was capable and experienced, entrusted him to take care of the situation. When he returned to work the next day, Dylan received an email from his customer. The message stated that, if the chip's manufacturing methodology was not disclosed, the customer would cease further investments in the product.

Shocked, Dylan believed that if the customer could not abide by the NDA, he should tell the contract should be broken off. However, doing so would mean losing a significant amount of profit they had intended on garnering from selling the chip. On the other hand, sharing confidential information with his customer could cause negative repercussions, especially if his company were to discover the legal breach.

Although it is highly unlikely that the extra chip information would be used by Dylan's client for malicious purposes, its disclosure could potentially affect his company's reputation, lead to mistrust in the company and compromise Dylan's position. What should Dylan do? **Evaluate the case from Utilitarian perspective**

2. Computer fraud

Suzanne works as a programmer in a corporation. Marcus, her supervisor, tells her to write a program to allow people to post entries directly to the company's accounting files ("the books"). Suzanne knows that ordinarily programs that affect the books involve several steps, all of which have to balance. She realizes that with the new program, it will be possible for one person to make changes to crucial amounts, and there will be no way to trace who made these changes, with what justification, or when.

Suzanne raises these concerns to Marcus, who tells her not to be concerned, that her job is simply to write the programs as he specifies. He says that he is aware of the potential misuse of these programs, but he justifies his request by noting that periodically a figure is mistakenly entered in the books and the company needs a way to correct the inaccurate figure.

If Suzanne writes this program, she might be an accomplice to fraud. If she complains to Marcus's superior, Marcus or the superior might reprimand or fire her as a troublemaker. If she refuses to write the program, Marcus can clearly fire her for failing to carry out an assigned task. We do not even know that the program is desired for fraudulent purposes; Marcus suggests an explanation that is not fraudulent.

She might write the program but insert extra code that creates a secret log of when the program was run, by whom, and what changes were made. This extra file could provide evidence of fraud, or it might cause trouble for Suzanne if there is no fraud, but Marcus discovers the secret log.

Consider the following questions (**respond to all questions from Utilitarian perspective**):

- a) Is a programmer responsible for the programs they write? Is a programmer responsible for the results of those programs? (In contemplating this question, suppose the program were to adjust dosage in a computer-controlled medical application, and Marcus's request were for a way to override the program controls to cause a lethal dosage. Would Suzanne then be responsible for the results of the program?)
- b) What degree of personal risk (such as possible firing) is an employee obliged to accept for opposing an action he or she thinks is improper?
- c) Would a program to manipulate the books as described here ever be justified? If so, in what circumstances would it be justified?

3. Hacking case

Miranda is a computer security consultant; she enjoys the challenge of finding and fixing security vulnerabilities. Independently wealthy, she does not need to work, so she has ample spare time in which to test the security of systems.

In her spare time, Miranda does three things: **First, she aggressively attacks commercial products for vulnerabilities. She is quite proud of the tools and approach she has developed, and she is quite successful at finding flaws. Second, she probes accessible systems on the Internet, and when she finds vulnerable sites, she contacts the owners to offer her services repairing the problems. Finally, she is a strong believer in high-quality pastry, and she plants small programs to slow performance in the web sites of pastry shops that do not use enough butter in their pastries. Let us examine these three actions in order (consider all the following points from Utilitarian perspective):**

- A) Who is helped or hurt by finding and publicizing flaws in products? Relevant parties are attackers, the vulnerability finder, the vendor, and the using public. Are the interests of the vendor (financial, public relations) less important than the interests of users to have secure products? Should Miranda disclose the security vulnerabilities to the users (full discloser? Partial discloser or no discloser?)?
- B) Is Miranda a “white hat” hacker or a “black hat” hacker?
- C) Is Miranda’s interference with the performance of the website of pastry shops that don’t use enough butter morally permissible?